A Non-Linear Economic Barrier Jump Model: Drawdown-Dependent Jump Risk in the S&P 500

L. D. Visconte

October 2026

Abstract

Standard jump-diffusion models treat jump arrivals as independent of market conditions, yet crashes tend to arrive when prices are already under stress. This paper proposes a non-linear economic barrier jump model: an extension of the Bates model in which both jump intensity and mean jump size depend, through a Gaussian kernel, on the distance between the asset price and an economic barrier (B). In its testable form the barrier is set at 80% of the trailing 252-day high and the kernel is one-sided, so jump risk rises as prices fall toward the barrier. Estimated on 2,519 daily S&P 500 log returns (September 2016–September 2026) and evaluated with the same PIT-residual framework used for the established extensions, the model achieves the highest log-likelihood (8,468.1), the best AIC of all six models tested, and the best tail fit (Jarque-Bera 48.2), and it is noticeably less surprised than Bates on the worst days near the barrier. However, the gain over Bates does not cover BIC's [Sch78] penalty for three added parameters, volatility dynamics are no better, and with only 73 trading days at or below the barrier the data cannot separate whether barrier-driven risk arrives as more frequent or as larger jumps.

1 Supported Extension

1.1 General Form of a Non-Linear Economic Barrier Jump Diffusion Model

In jump diffusion, jumps operate as random events; however, jump intensity also arrives. This is an issue since when assets experience black swan events, their jump intensities often increase, and are no longer independent of current asset price or the barrier or black swan event formulated as an enemy barrier (B).

Drawing from Bates SDE [Bat96] concerning asset price, we can reformulate the drift term and jump intensity $\lambda(\cdot)$ to be dependent on current price S_t and on the magnitude of an economic barrier B as such:

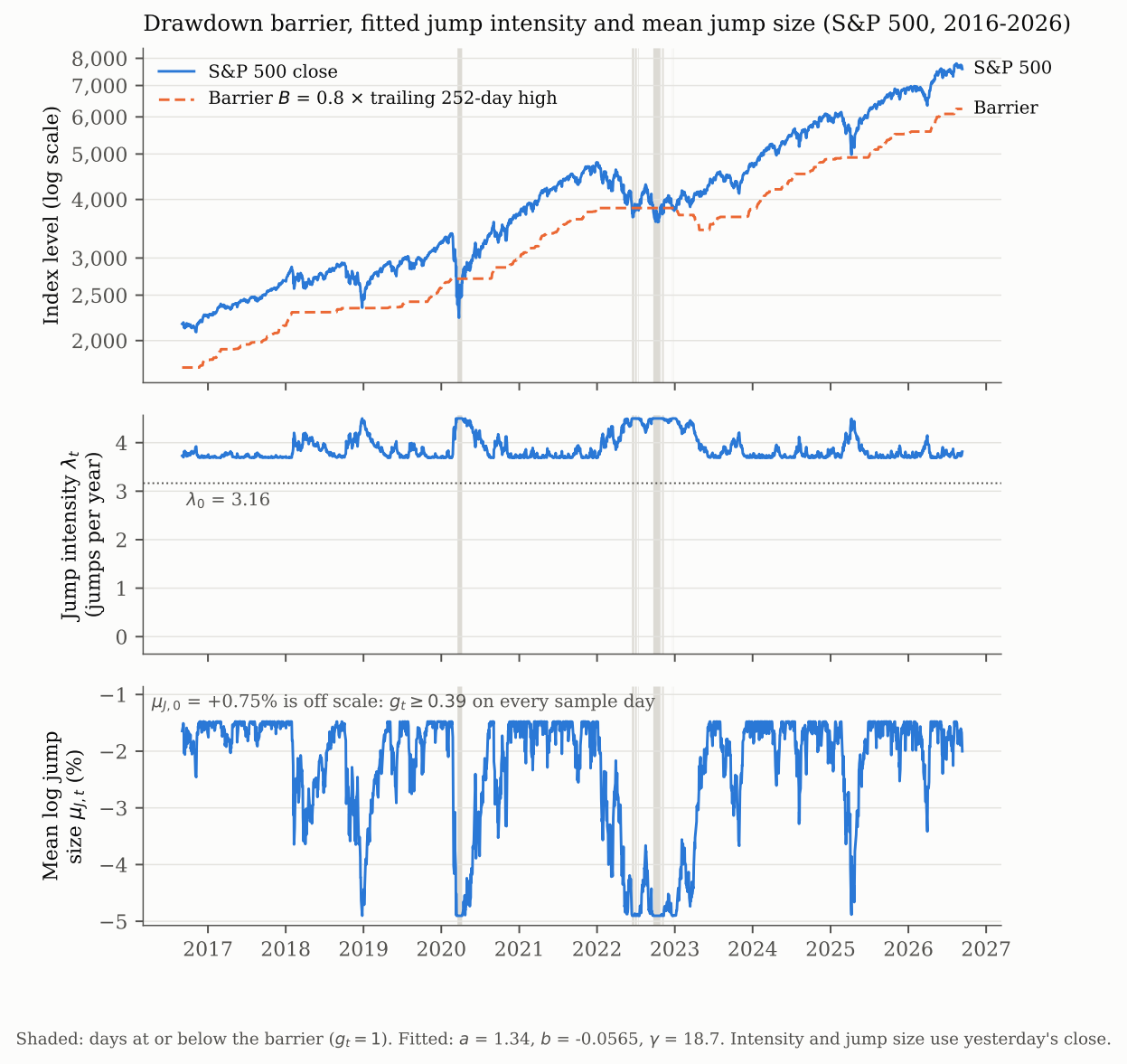
$$dS_t = \left(\mu - \lambda(S_t)\bar{k}\right) S_t\, dt + \sqrt{v_t} S_t\, dW_t^S + (J-1) S_t\, dN_t$$

where $\bar{k}$= E[J - 1]
where W_t^S & v_t follow Heston volatility dynamics [Hes93]
where N_t is a typical Poisson process driving jumps [Mer76]
where $\lambda(S_t)$ is a non-linear jump intensity function defined in proportion to B[1]:

$$\lambda(S_t) = \lambda_0 + a \exp\left(-\gamma(S_t - B)^2\right)$$

where $B = f(X_t)$ where X_t is an economic state vector
where λ_0 operates as the jump intensity in standard jump diffusion
where γ (gamma) is the "kernel" bandwidth sensitivity parameter
where a is a barrier induced intensity increment.

[1]Letting jump intensity depend on the state of the process has precedent: [DPS00] allow jump intensity to be affine in the state vector, [CL06] make default intensity a function of the stock price level, and [CJO12] estimate time-varying jump intensities on S&P 500 returns. The squared-exponential form used here is the Gaussian kernel of kernel density estimation [Ros56, Par62], with γ playing the role of an inverse bandwidth.



Shaded: days at or below the barrier ($g_t = 1$). Fitted: a = 1.34, b = -0.0565, γ = 18.7. Intensity and jump size use yesterday's close.

1.2 Testable Form of a Non-Linear Economic Barrier Jump Diffusion Model

This system will be tested during log difference/distance meaning:

$$\lambda(S_t) = \lambda_0 + a \exp\left(-\gamma([\ln(S_t) - \ln(B)])^2\right)$$

One sided form:

$$\lambda_t = \lambda_0 + a \exp\left(-\gamma\left[\max(0, \ln S_{t-1} - \ln B_{t-1})\right]^2\right) \tag{1}$$

Jump frequency & jump size changing total formulation:

$$dS_t = (\mu - \lambda_t \bar{\kappa}_t) S_t dt + \sqrt{v_t} S_t dW_t^S + (J_t - 1) S_t dN_t \tag{2}$$

where:

$$\bar{\kappa}_t = \exp\left(\mu_{J,t} + \frac{1}{2}\sigma_J^2\right) - 1 \tag{3}$$

where:

$$\lambda_t = \lambda_0 + a f_t \tag{4}$$

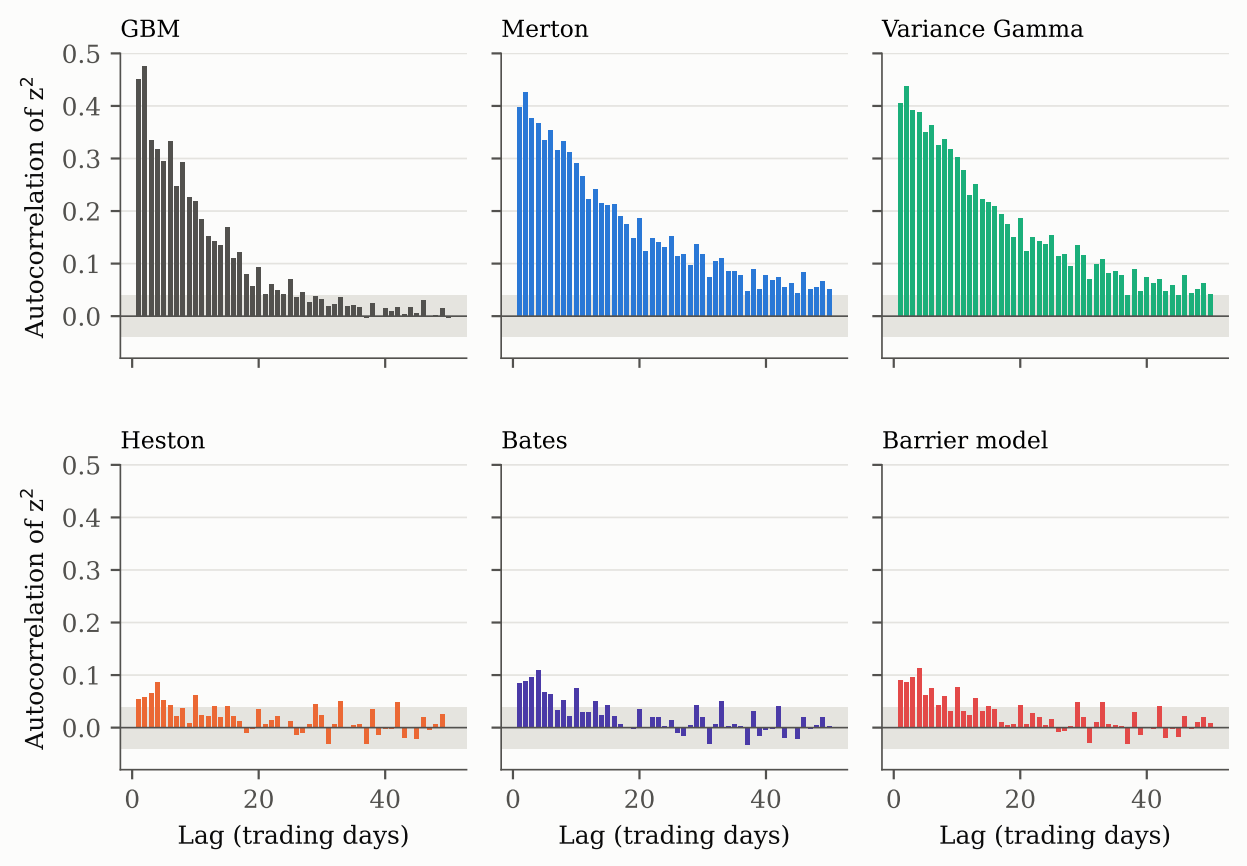
$$\mu_{J,t} = \mu_{J,0} + b f_t \tag{5}$$

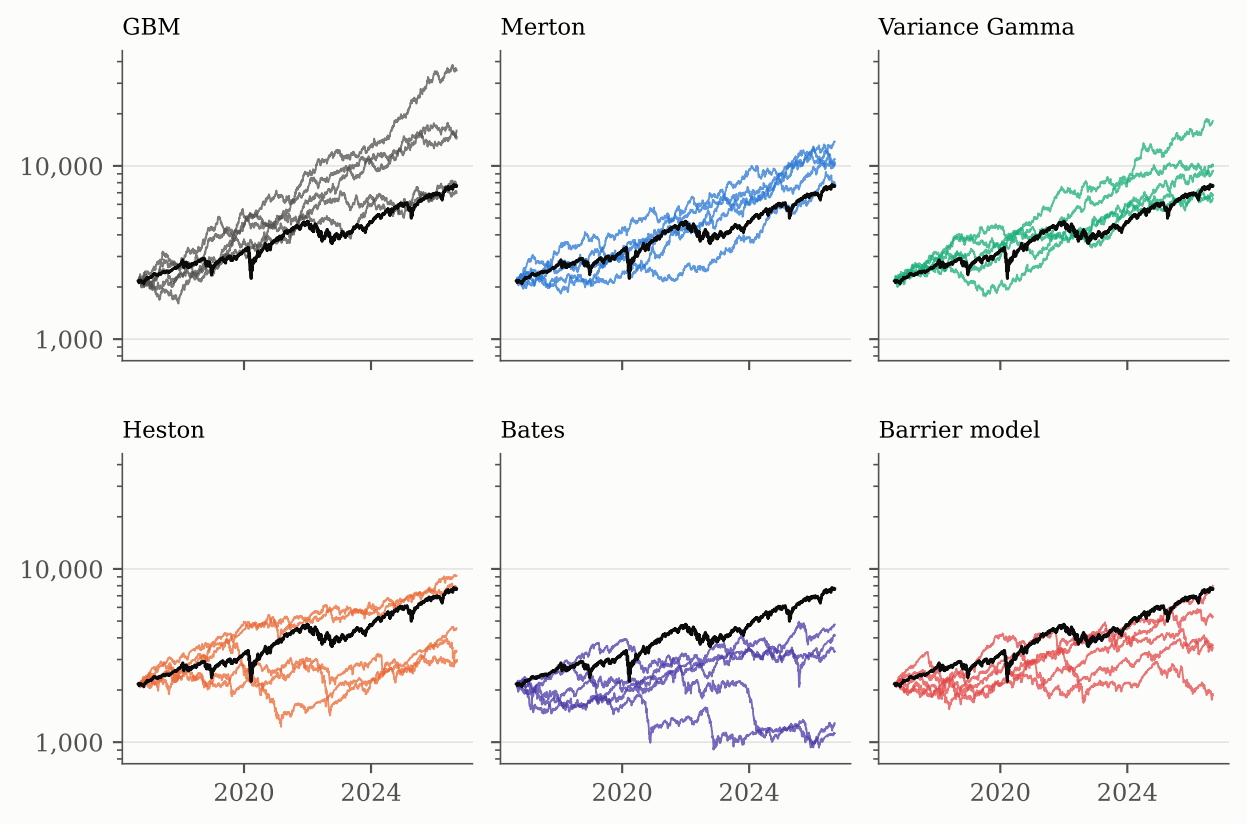
with:

$$f_t = \exp\left(-\gamma \left[\max(0, \ln S_{t-1} - \ln B_{t-1})\right]^2\right) \tag{6}$$

Where:

- a is the B-induced jump chance increment
- b is the B-induced jump size increment
- $B = \frac{8}{10} \times$ trailing 252 day high, moving peak from Sept 2015 (then smooth disaster)
- $a > 0$, $\gamma > 0$, $b < 0 \rightarrow$ stress makes jumps more negative (more intensity)




GBM
Merton
Variance Gamma
Heston
Bates
Barrier model
10,000
1,000
2020
2024

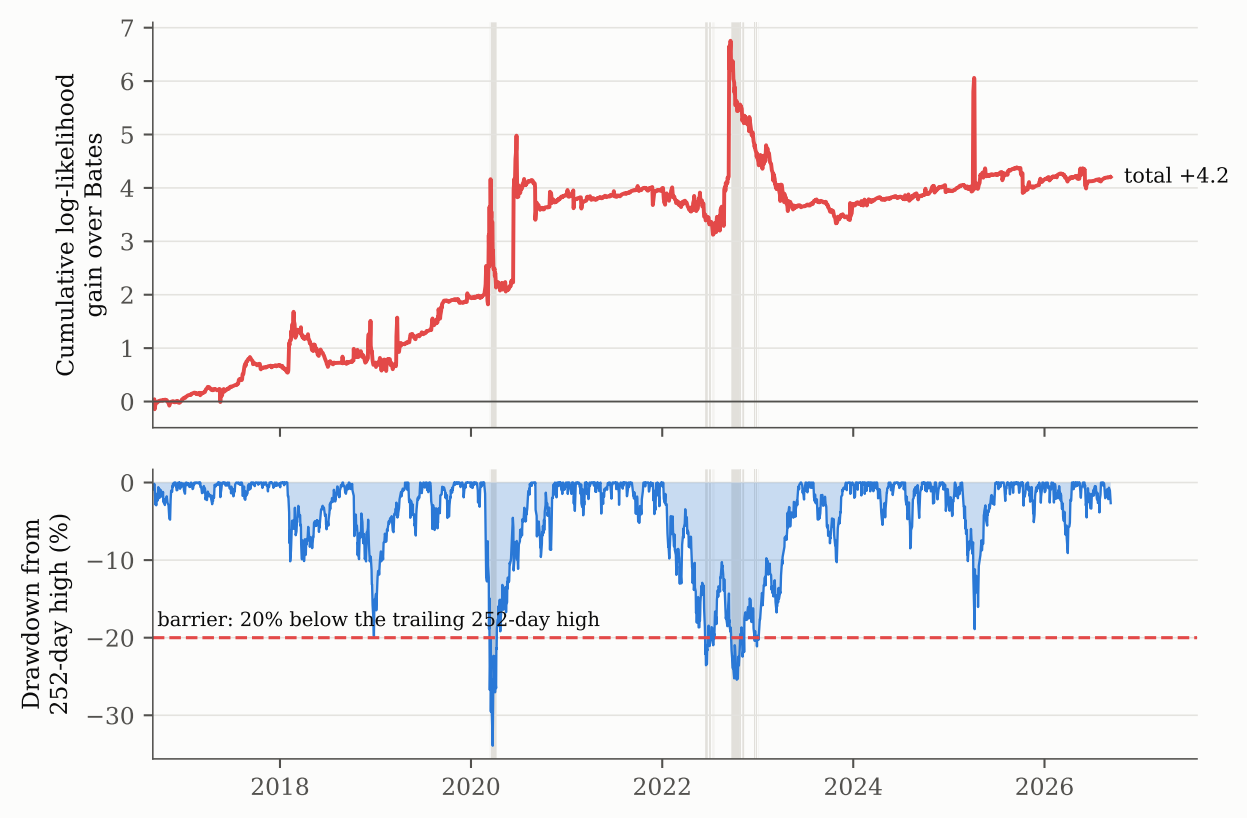
2 Supported Extension Testing

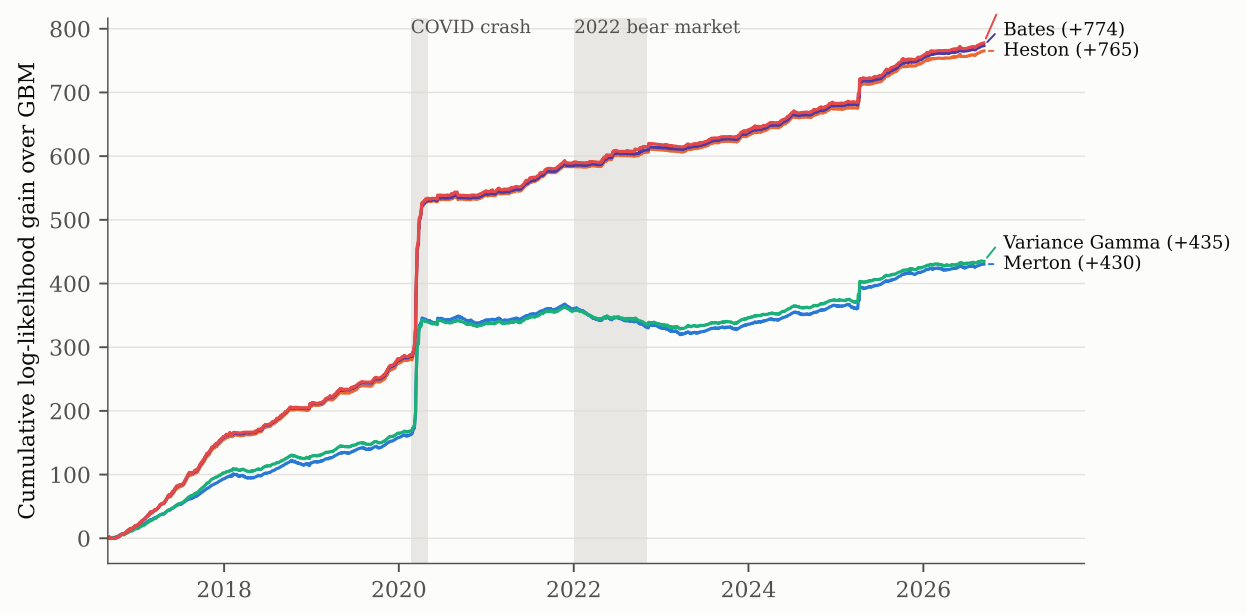
Table 1: Specification diagnostics – Supported Extension (Drawdown-Barrier Bates) (S&P 500, 2016-09-01 to 2026-09-11, $N = 2{,}519$)

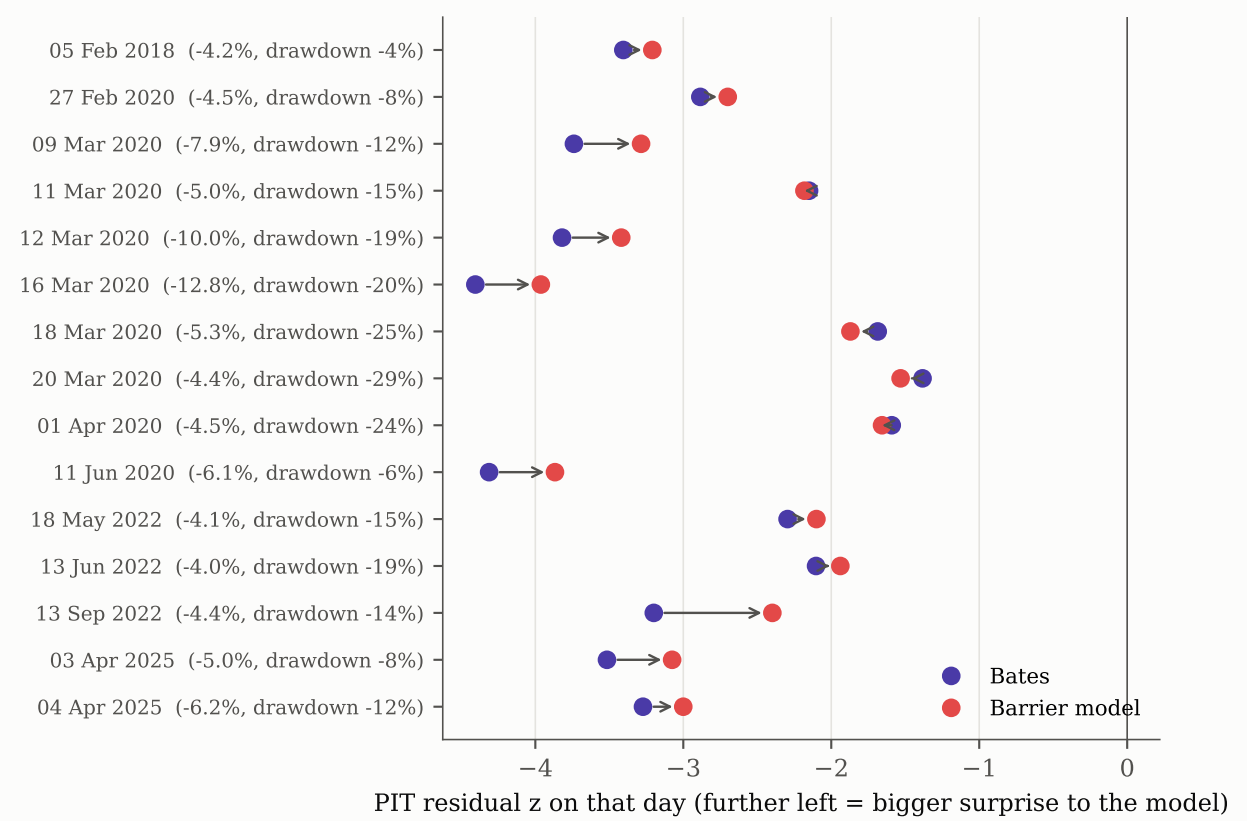
Statistic	GBM (Part 4)	Bates	Barrier model	Reference	p-value (Barrier model)
Panel A: Distribution of PIT residuals					
Skewness	−0.678	−0.367	−0.283	0	
Kurtosis	19.69	3.521	3.373	3	
Jarque-Bera	29,431.8	84.9	48.2	$\chi^2(2)$	<0.001
Panel B: Serial dependence (Ljung-Box)					
$Q_{\text{raw}}(10)$	255.7	20.5	21.6	$\chi^2(10)$	0.017
$Q_{\text{raw}}(100)$	427.1	116.8	117.3	$\chi^2(100)$	0.114
$Q_{\text{sq}}(10)$	2,741.2	138.2	150.8	$\chi^2(10)$	<0.001
$Q_{\text{sq}}(100)$	3,281.7	280.4	287.7	$\chi^2(100)$	<0.001
Panel C: Variance ratios					
VR(2), cumulated returns / z	0.856	0.959	0.955	1	0.042
VR(20), cumulated returns / z	0.806	0.871	0.852	1	0.169
VR(10), realized vol 21d	2.579	1.225	1.220	1	0.001 (MC)
VR(21), realized vol 63d	5.458	1.649	1.715	1	<0.001 (MC)
Panel D: Level statistics, model-implied vs. S&P 500 (1,000 simulated paths)					
ADF, log price	−0.86	−1.34	−1.31 [−2.62, 0.21]	−0.48 (S&P)	0.346 (MC)
ADF, realized vol 21d	−6.79	−4.96	−4.90 [−5.87, −3.89]	−5.08 (S&P)	0.756 (MC)
OU half-life, RV 21d (days)	13.2	58.1	45.5 [33.3, 61.9]	100.5 (S&P)	<0.001 (MC)
OU half-life, RV 63d (days)	40.8	279.1	217.1 [141.9, 352.2]	428.0 (S&P)	0.046 (MC)
Panel E: Fit					
Log-likelihood	7,692.8	8,463.7	8,468.1		
AIC	−15,381.7	−16,911.4	−16,914.3	lower is better	
BIC	−15,370.0	−16,864.7	−16,850.1	lower is better	
Parameters (k)	2	8	11		

Notes: 'GBM (Part 4)' repeats the Section 4 statistic (raw returns are GBM's PIT residuals); 'Barrier model' is the same statistic on this model's PIT residuals $z_t = \Phi^{-1}(F_t(r_t))$, i.i.d. $N(0,1)$ under correct specification; 'Reference' is the value a correct model should produce. p-values refer to the Barrier model column: asymptotic, except (MC) = Monte Carlo (Panel C realized-vol rows against 2,000 i.i.d. $N(0,1)$ series; Panel D: S&P value against 1,000 paths simulated from the fitted model). Variance-ratio p-values use the heteroskedasticity-robust z^*. Panel D: GBM and model columns are simulated medians, model 90% band in brackets; OU p-values are computed on the AR(1) coefficient b. Estimates (annualised, under $\mathbb{P}$): $\mu = 0.06349$, $\kappa = 7.859$, $\theta = 0.02747$, $\xi = 0.5714$, $\rho = -0.7298$, $\lambda_0 = 3.165$, $\mu_{J,0} = 0.007504$, $\sigma_J = 0.0009119$, $a = 1.335$, $b = -0.05653$, $\gamma = 18.7$. Bates column: the nested model ($a = b = 0$); Panel D shows its simulated median. Barrier $B = 0.8\times$ trailing 252-day high, one-sided kernel $g_t = \exp(-\gamma[\max(0, \ln S_{t-1} - \ln B_{t-1})]^2)$, $\lambda_t = \lambda_0 + a\, g_t$, $\mu_{J,t} = \mu_{J,0} + b\, g_t$. $g_t \geq 0.39$ on every sample day, so in sample λ_t spans 3.7–4.5 jumps per year and $\mu_{J,t}$ spans −1.5% to −4.9%; λ_0 and $\mu_{J,0}$ are extrapolations. Three starts reached the same likelihood (within 0.2) with different frequency/size splits (e.g. $\lambda_0 \approx 0$, $a \approx 30$, $b \approx 0$): barrier-driven jump risk is supported, its split between frequency and size is not identified. σ_J at its lower bound.

The barrier model achieves the highest log-likelihood (8,468.1) and the best AIC [Aka74] of all six models, and the best tail fit: JB falls to 48.2 and skewness to −0.28. On the worst days near the barrier — for example 16 March 2020 — it is noticeably less surprised than Bates. Its gains are, however, modest: the 4.4-point improvement over Bates does not cover BIC's penalty for three added parameters, and volatility dynamics are no better than Bates's. Two limitations matter for interpretation. First, the fitted kernel never fully switches off ($g_t \geq 0.39$), so the barrier acts partly as a general drawdown effect. Second, only 73 trading days fall at or below the barrier, so the data support barrier-driven jump risk but cannot separate whether it arrives as more frequent or larger jumps.


Cumulative log-likelihood gain over Bates
total +4.2
Drawdown from 252-day high (%)
barrier: 20% below the trailing 252-day high
2018
2020
2022
2024
2026


Cumulative log-likelihood gain over GBM
COVID crash
2022 bear market
Bates (+774)
Heston (+765)
Variance Gamma (+435)
Merton (+430)
2018
2020
2022
2024
2026



3 Conclusion

In conclusion, Geometric Brownian Motion fails on the S&P 500 testing in three ways: fat tails, volatility clustering, and volatility shocks. Jumps (Merton, Variance Gamma) fix the issue of fat tailing, but not the volatility dynamics; stochastic volatility (Heston) fixes most of the second issue, and combining both (Bates) trades a better shape for slightly worse clustering. No model reproduces the full persistence of volatility. The barrier model shows that tying jump risk to an economic barrier improves the fit where markets are most stressed, though the applications of this model are limited by how rarely such stress occurs. These cumulative limitations give rise to the natural next steps of longer samples containing more crises (or periods of market stress), and a barrier built from macroeconomic variables rather than in relation to asset prices.

References

[Aka74] H. Akaike, *A new look at the statistical model identification*, IEEE Trans. Automat. Control **19** (1974), no. 6, 716–723.

[Bat96] D. S. Bates, *Jumps and stochastic volatility: exchange rate processes implicit in Deutsche Mark options*, Rev. Financ. Stud. **9** (1996), no. 1, 69–107.

[CJO12] P. Christoffersen, K. Jacobs, and C. Ornthanalai, *Dynamic jump intensities and risk premiums: evidence from S&P500 returns and options*, J. Financ. Econ. **106** (2012), no. 3, 447–472.

[CL06] P. Carr and V. Linetsky, *A jump to default extended CEV model: an application of Bessel processes*, Finance Stoch. **10** (2006), no. 3, 303–330.

[DPS00] D. Duffie, J. Pan, and K. Singleton, *Transform analysis and asset pricing for affine jump-diffusions*, Econometrica **68** (2000), no. 6, 1343–1376.

[Hes93] S. L. Heston, *A closed-form solution for options with stochastic volatility with applications to bond and currency options*, Rev. Financ. Stud. **6** (1993), no. 2, 327–343.

[Mer76] R. C. Merton, *Option pricing when underlying stock returns are discontinuous*, J. Financ. Econ. **3** (1976), no. 1–2, 125–144.

[Par62] E. Parzen, *On estimation of a probability density function and mode*, Ann. Math. Statist. **33** (1962), no. 3, 1065–1076.

[Ros56] M. Rosenblatt, *Remarks on some nonparametric estimates of a density function*, Ann. Math. Statist. **27** (1956), no. 3, 832–837.

[Sch78] G. Schwarz, *Estimating the dimension of a model*, Ann. Statist. **6** (1978), no. 2, 461–464.